

Assignment 1 - Client-side Development

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Web Application Development (COMP5347)

School of Computer Science

Assignment 1 - Client-side Development 客户端开发

Goals and Background

Goal of the assignment

The primary goal of this assignment is to demonstrate your fundamental understanding of the core client-side technologies (HTML, CSS, and JavaScript) and your ability to create well-structured HTML pages with effective styling and interactive features.

展示HTML,CSS,JavaScript技能和创建一个结构良好的带有有效的样式和交互功能的HTML页面。

Background of the assignment

B2YBooks, a book selling company, is implementing a digital transformation strategy that includes developing a web store for online book sales.

背景: 图书销售公司B2YBooks正在实施数字化转型战略, 包括**开发在线图书销售网店**。

As a front-end web developer, your task is to create the main web page with key functionalities and features defined in the first development iteration. To assist you, they have provided a *starter file set* (see Resources below), which includes a complete CSS file and a skeleton HTML file.

开发一个**主网页**, 其中包含在第一次开发迭代中定义的关键功能和特性。提供了一个起始文件集 (请参阅下面的参考资料), 其中包括一个完整的CSS文件和一个骨架超文本标记语言文件。

Your task is to develop the main web page of B2YBooks online store while adhering to the following requirements without using any third-party CSS frameworks or JavaScript libraries.

您的任务是在不使用任何第三方CSS框架或JavaScript库的情况下, 遵循以下要求开发B2YBooks在线商店的主网页。

Main Requirements

The following functionalities **MUST** be implemented on your Web page:

Requirement	Description
Main Page	The main page should prominently display the website name and key details for each book, including a thumbnail image, title, author, publication year, star rating, price, publisher, and category (see the enclosed figures under Resources for reference). The star rating can be positioned either inline with the title or in a separate column (e.g., a "Ratings" column following the "Title" column). 主页应突出显示每本书的网站名称和关键详细信息，包括缩略图、标题、作者、出版年份、星级、价格、出版商和类别（请参阅参考资料下的随附图）。星级可以与标题内联或单独放置（例如，“标题”列后面的“评级”列）。
Dynamic Data 动态数据	The page must dynamically load the list of book items from the provided JSON file (included in the starter file set; see the Resources section below). Instructions on how to read the JSON file will be provided later in this assignment description. Directly copying the contents of the JSON file into your JavaScript code is not permitted. 页面必须从提供的JSON文件（包含在起始文件集中；请参阅下面的参考资料部分）中动态加载书籍项目列表。关于如何阅读JSON文件的说明将在本作业说明的后面提供。不允许将JSON文件的内容直接复制到JavaScript代码中。
Search Function 查询功能	End users should be able to search for books by title. Matching rows will be highlighted with a background colour of your choice, but the search will not affect the displayed list—it only changes the background colour of rows containing the search term (which can be a word, multiple words, or even a single letter). 最终用户应该能够按标题搜索书籍。匹配行将使用您选择的背景颜色突出显示，但搜索不会影响显示的列表——它只会更改包含搜索词（可以是一个单词、多个单词，甚至一个字母）的行的背景颜色。

Filter Function 过滤功能	Users should be able to filter books by category, displaying only those that match the selected category. The category selection should be implemented as a dropdown menu, including an extra category not currently available in the bookstore for boundary testing. Users should always have the option to reset the filter and return to the default view, displaying all books regardless of category. 用户应该能够按类别过滤书籍，只显示与所选类别匹配的书籍。类别选择应实现为下拉框，包括书店中当前不可用的额外类别以进行边界测试。用户应该始终可以选择重置过滤器并返回默认视图，显示所有书籍，而不管类别如何。
Search and Filter Function together 合并查找和过滤	The Search and Filter functions should work together, combining their results dynamically. 搜索和过滤函数应该协同工作，动态组合它们的结果。
Add to Cart Function 添加到购物车	Users should be able to select books and add them to the Shopping Cart using checkboxes and an "Add to Cart" button. However, only one book can be selected at a time—functioning like a radio button but with a checkmark. 用户应该能够使用复选框和“添加到购物车”按钮选择书籍并将其添加到购物车。然而，一次只能选择一本书——功能类似于单选按钮，但带有复选标记。
Add to Cart Quantity Function 添加到购物车的数量	When a user selects a book and clicks the "Add to Cart" button, a quantity input box should appear (either as a popup or a text box) asking for the desired quantity. Once a book is added to the cart, the selection checkbox should be cleared, but the book should remain visible in the list. 当用户选择一本书并单击“添加到购物车”按钮时，应该会出现一个数量输入框（弹出窗口或输入框），询问所需的数量。一旦将一本书添加到购物车，选择复选框应该被清除，但这本书应该在列表中保持可见。
Reset Cart Functions 重置购物车	Users can reset the shopping cart using a "Reset the Cart" button. Before clearing the cart, they should receive a confirmation prompt:

	<i>"Are you sure you want to reset the cart?" with "OK" and "Cancel" options.</i> 用户可以使用“重置购物车”按钮重置购物车。在清除购物车之前，他们应该会收到一个确认提示： “您确定要重置购物车吗？”带有“确定”和“取消”选项。
Shopping Cart 购物车	The Shopping Cart must always display the correct total number of books based on user actions, reflecting the total quantity of added books rather than just the number of unique book titles. 购物车必须始终根据用户操作显示正确的书籍总数，反映添加书籍的总数，而不仅仅是唯一书名的数量。
"Dark mode" function 深色模式功能	the page should include a "Dark Mode" toggle (positioned at the top of the page), allowing users to switch to a dark-themed interface. When enabled, this mode should adjust the page's background, text color, highlight colors, and any other relevant UI elements to use darker shades (e.g., greys and other dark tones) for better readability in low-light conditions. 该页面应包括一个“深色模式”切换（位于页面顶部），允许用户切换到深色主题界面。启用后，此模式应调整页面的背景、文本颜色、突出显示颜色和任何其他相关的用户界面元素，以在弱光条件下使用较暗的色调（例如灰色和其他深色调）以获得更好的易读性。

Design and Implementation Requirements

Attention Please

- The web design implementation **MUST NOT** use any external resources such as frameworks or libraries
- You **MUST** develop everything from scratch using only plain JS, HTML, and CSS.
- Failure to comply with this requirement will result in a score of ZERO marks.

网页设计实现不得使用任何外部资源，例如框架或库

您必须从头开始开发所有内容，只使用纯JS、超文本标记语言和CSS。

不遵守此要求将导致零分。

The online bookstore data and required thumbnail images are provided in the starter file set (see Resources below). While your implementation does not need to exactly match the design mock-ups in Figure 1, you are encouraged to customise the provided CSS file to refine the colour scheme, borders, and text styles, allowing for a slightly different but well- structured design.

Additionally, you **must** account for boundary cases and implement appropriate UI responses to **notify/alert** users of invalid inputs or unexpected outputs. Example of boundary cases include, but are **not limited** to:

- A search term that does not match any characters.
- A selected filter that does not correspond to any characters.

联机书店数据和所需的缩略图在起始文件集中提供（请参阅下面的参考资料）。但是您的实现不需要完全匹配图1中的设计模型，我们鼓励您自定义提供的CSS文件，以优化配色方案、边框和文本样式，从而实现略有不同但结构良好的设计。

此外，您必须考虑边界情况并实施适当的UI响应，以通知/提醒用户无效输入或意外输出。边界情况的示例包括但不限于：

不匹配任何字符的搜索词。

不对应任何字符的选定过滤器。

Code Structure and Best Practices:

- Ensure that your HTML, CSS rules and JavaScript code are kept in separate files. **Inline styles or JavaScript code are not allowed**.
- Make sure that your HTML and CSS files are “clean” and easy to maintain. Avoid unnecessary, automatically generated elements, such as scattered tags with fragmented styles.
- Use of jQuery or any other JavaScript libraries is strictly prohibited. Similarly, Bootstrap or any other CSS frameworks/libraries are **NOT** allowed.
- Add clear and meaningful comments throughout your code to improve readability and assist the marker in understanding your implementation.

代码结构和最佳实践：

确保您的超文本标记语言、CSS规则和JavaScript代码**保存在单独的文件中。不允许内联样式或JavaScript代码。**

确保你的超文本标记语言和CSS文件是“干净的”并且易于维护。避免不必要的、自动生成的元素，例如带有碎片化样式的分散标记。

严禁使用jQuery或任何其他JavaScript库。同样，不允许使用Bootstrap或任何其他CSS框架/库。

在整个代码中添加清晰而有意义的**注释**，以提高易读性并帮助标记理解您的实现。

Deliverables and Submission Guidelines

- **Submit a zip file which should contain**
 - All CSS, JavaScript, HTML files required to view the website design; **AND**
 - A readme file that contains instructions to your marker on how to run your application and anything that you would like to draw your marker's attention for your assignment; **AND**
 - The json file and the images for which you need to make sure their folder path are correct (ensure all paths to resources and images are relative, **NOT** absolute).
- **Attend the Oral Assessment on Week 5**
 - Details could be found [here](#).
- **Attention Please**
 - The format of the submitted file **MUST** be .zip rather than .7z or .rar
 - All the submitted code will be executed so all the Web page functionalities can be tested and assessed following the assignment requirements (above) and marking criteria listed below.
 - The zip file should include Unikey in the file name. For example, *asabcd1234_Assignment1.zip* where abcd1234 is your Unikey.
 - A software system will be used to detect software code similarity and allegations will be further investigated by the University's integrity office.

可交付成果和提交指南

提交一个应该包含的**zip文件**

查看网站设计所需的**所有CSS、JavaScript、超文本标记语言文件**；和

一个自述文件，其中包含有关如何运行应用程序以及您希望在分配任务时引起标记注意的任何内容的标记说明；和

您需要确保其文件夹路径正确的json文件和图像（确保资源和图像的所有路径都是相对的，而不是绝对的）。

参加第5周的**口头评估**

这里可以找到详细信息。

请注意：

提交文件的格式必须是 **.zip** 而不是 .7z 或 .rar

所有提交的代码都将被执行，以便可以按照分配要求（上面）和下面列出的标记标准测试和评估所有网页功能。

zip文件应该**在文件名中包含Unikey**。例如，asabcd1234_Assignment1.zip其中abcd1234是您的Unikey。

软件系统将用于检测软件代码相似性，指控将由大学诚信办公室进一步调查。

Marking Guide

- 8 marks are awarded for successful implementation of the features.
- 1 mark is awarded for code quality
- 1 mark is awarded for using the JSON data successfully
- 5 marks are awarded for answering **all** questions correctly from your Tutor during the Oral Assessment component.
- Your marks will be scaled according to how closely it matches the overall layout and theme of the mock-ups. For example, if your design has some minor issues with matching the mock-up layout or colour, your mark may be capped at 90%. Major design issues will cap your mark at 70%.

分数指南

成功实施功能将获得8分。

代码质量获得1分

成功使用JSON数据可获得1分

在口头评估环节中，正确回答导师的所有问题将获得5分。

您的分数将根据其与模型的整体布局和主题的匹配程度进行缩放。例如，如果您的设计在与模型布局或颜色匹配方面存在一些小问题，您的分数可能会被限制在90%。重大设计问题将使您的分数限制在70%。

Resources

- [Starter File Set](#) [Download Starter File Set](#) (HTML skeleton, JSON data and CSS files)

- [Design Mock-up Download Design Mock-up](#) (An example of the requirements and design of B2YBooks web page)

Assignment Oral Assessment

Each student will have to do an oral assessment and demo of their submitted assignment to their tutor during week 5's labs. Every student should be able to explain and answer code-related questions as part of their assignment work. You will be asked a series of questions from your Tutor during class to check your understanding of your work. These questions will be related to the implementation of the design. This component will take around 5 minutes per student.

每个学生**在第五周的实验中都必须对他们提交的作业进行口头评估和演示。每个学生都应该能够解释和回答与代码相关的问题**，作为他们作业工作的一部分。在课堂上，你的导师会问你一系列问题，以检查你对工作的理解。这些问题将与设计的实施有关。每个学生需要大约5分钟的时间来完成这个部分。

Supporting Material to Load a JSON File 加载JSON文件！

Before being able to read the json file, you will need to setup a simple HTTP server first. You can either use PHP or Node.js to run the simple HTTP server. Please select one of these options:

- **Using PHP** : For Mac OS or linux users, your operating system usually has PHP installed so you do not have to install it. For windows users, you will need to install PHP since it is not available by default. After having PHP installed, you first open your command line or terminal and navigate to your project directory by using "cd" command. Afterwards, you run this command: "`php -S localhost:8000`" and it should say "*Listening on <http://localhost:8000>*". By running this command you can open your project from your browser using this address: "<http://localhost:8000>".
- **Using Node.js** : You will be using it for upcoming tutorials and you can use it to work on your second assignment, so installing Nodejs can serve other purposes. They provide a UI installer so it is easy to install, you can go to this link: <https://nodejs.org/en/download/> [Links to an external site.](#) . After installing Node.js, open your terminal (mac/linux) or command prompt (windows) and run this command: "`npm install http-server -g`" (put sudo before npm if you are mac/linux users). If http-server installation is complete, go to your project directory by using "`cd`" command and run command "`http-server`". It will provide you url(s) to access your project from your browser, you usually can access your project from this address: "<http://127.0.0.1:8080>".

Having one of above options ready, you have to insert this function into your javascript file:

```
1  function getJsonObject(path, success, error) {
```

```

2     var xhr = new XMLHttpRequest();
3     xhr.onreadystatechange = function() {
4         if (xhr.readyState === XMLHttpRequest.DONE) {
5             if (xhr.status === 200) {
6                 if (success) success(JSON.parse(xhr.responseText));
7             } else {
8                 if (error) error(xhr);
9             }
10        }
11    };
12    xhr.open("GET", path, true);
13    xhr.send();
14 }

```

Finally, to read the json file, you can call the function by something like this inside your *window.onload* method:

```

1  characterList = []; // character list container
2  getJsonObject('data.json',
3      function(data) {
4          characterList = data; // store the character list into characterList
5          console.log(characterList); // print it into console (developer tools)
6          console.log(characterList[0]); // print the first character object to
           the console
7          // here you can call methods to load or refresh the page
8          // loadCharacters() or refreshPage()
9      },
10     function(xhr) { console.error(xhr); }
11 );

```

When you have added these lines of code into your JavaScript file, you can open your browser's developer tool and check the console output. Make sure you hard reload/refresh the page after adding these lines (windows/linux: " *Ctrl+F5*" and mac: " *⌘+⇧+R*")

Academic integrity

While the University is aware that the vast majority of students and staff act ethically and honestly, it is opposed to and will not tolerate academic integrity breaches and will treat all allegations seriously.

Further information on academic integrity, and the resources available to all students can be found on the academic integrity pages on the current students website:

<https://sydney.edu.au/students/academic-integrity.html> .

You may only use generative artificial intelligence (AI) and automated writing tools in assessment tasks if you are permitted to by your unit coordinator. If you do use these tools, you must acknowledge this in your work, either in a footnote or an acknowledgement section. For information on acknowledging AI please refer to the guidance in the [AI in Education Canvas site](#) .

We use Turnitin, which includes AI detection, to detect potential instances of plagiarism or other forms of academic integrity breach. If such matches indicate evidence of plagiarism or other forms of academic integrity breaches, your teacher is required to report your work for further investigation.

Further information for on research integrity and ethics for postgraduate research students and students undertaking research-focussed coursework such as Honours and capstone research projects can be also be found on the current students website:

<https://sydney.edu.au/students/research-integrity-ethics.html> .

Compliance statement

In submitting this work, I acknowledge I have understood the following:

- I have read and understood the University of Sydney's [Academic Integrity Policy](#) .
- The work is substantially my own and, where any parts of this work are not my own, I have indicated this by acknowledging the source of those parts of the work and clearly indicated any quoted text by quotation marks or indentation according to accepted style standards.
- I have acknowledged any assistance provided in preparing the work including the use of copy-editing, proof-reading, and automated writing and drawing tools (including artificial intelligence (AI), reference generators, translation software, grammar checkers, but not spell checkers).
- The work has not previously been submitted in part or in full for assessment in another unit unless I have been given permission by my unit of study coordinator to do so.
- The work will be submitted to similarity detection software (Turnitin) and a copy of the work will be retained in Turnitin's paper repository for future similarity checking.
- Engaging in plagiarism or academic dishonesty in coursework will, if detected, lead to the University commencing proceedings under the [Academic Integrity Policy](#). and the [Academic Integrity Procedures](#) .
- Engaging in plagiarism or academic dishonesty in research-focussed work will lead to the University commencing proceedings under the [Research Code of Conduct](#) and the [Academic Integrity Procedures](#) .

- Engaging another person to complete part or all of the submitted work will, if detected, lead to the University commencing proceedings against me for potential student misconduct under the [University of Sydney \(Student Discipline\) Rule](#) .

You are advised to keep copies of your assignment submission, drafts and any other research materials as evidence of your research and writing process. If you have used AI in the completion of your assignment, you should keep copies of the AI outputs.

建议你保留你提交的作业、草稿和任何其他研究材料的副本，作为你研究和写作过程的证据。如果你在完成作业时使用了AI，你应该保留AI产出的副本。

Criteria	Ratings	Pts
COMP5347_2024_Assignment_1_Rubric		
Criteria	Ratings	Pts
Main Page	0.5 pts Fully and correctly implemented The page must show the correct website name either at the top of the page or at any noticeable location 充分和正确实施 页面必须在页面顶部或任何明显位置显示正确的网站名称	/ 0.5 pts
Dynamic Data	1 pts Fully and correctly implemented with no issues in the content and layout All 10 books with their thumb, title, authors, year, star ratings, price, publisher and category are displayed completely. 完全正确地实施，内容和布局没有问题 所有10本书及其拇指、标题、作者、年份、星级、价格、出版商和类别都完整显示。	/ 1 pts
Main components	0.5 pts Fully and correctly implemented with all of the elements specified in the requirements Search input textbox, filter drop down menu, dark mode toggle and relevant buttons, e.g. search, filter, reset, “add to cart” and “reset the cart” button, cart icon and number indicator 使用需求中指定的所有元素完全且正确地实现	/ 0.5 pts

	搜索输入文本框、筛选下拉菜单、深色模式切换和相关按钮，例如搜索、筛选、重置、“加入购物车”和“重置购物车”按钮、购物车图标和数字指示器	
CSS Style	1 pts Fully and correctly implemented without missing styles The layout looks clear and user-friendly. It expresses the information in a readable way and the function buttons are recognisable. 完全正确地实现，不遗漏样式。布局看起来清晰且用户友好。它以可读的方式表达信息，功能按钮易于识别。 0.8 pts Not fully and/or correctly implemented Input buttons are not aligned, or the table is not delivered in a readable and user-friendly way. 输入按钮没有对齐，或者表格没有以可读性和用户友好的方式传递。	/ 1 pts
Search Feature	1 pts Fully and correctly implemented with the consideration of boundary cases The searched row is highlighted correctly. The search function works as expected. 充分且正确地实现，考虑边界情况搜索的行正确高亮。搜索功能按预期工作。	/ 1 pts
Filter category feature	1 pts Fully and correctly implemented with the consideration of boundary cases Category filter button will filter the books belonging to the selected category 在考虑边界情况的情况下完全正确地实现类别过滤器按钮将过滤属于所选类别的书籍	/ 1 pts
Successfully combine the result of search and filter feature	1 pts Fully and correctly implemented with the consideration of boundary cases The filter can be operated upon the output of the search and vice versa. 在考虑边界情况的情况下完全正确地实施过滤器可以根据搜索的输出进行操作，反之亦然。	/ 1 pts
Add to Cart feature	1 pts Fully and correctly implemented with the consideration of boundary cases	/ 1 pts

	The number of added to cart is reflected properly by considering the quantity of the added books. 在考虑边界case的情况下，充分且正确地执行 通过考虑添加的书籍数量，正确反映添加到购物车的数量。	
Reset the cart feature	0.5 pts Fully and correctly implemented with reasonable prompt box Cart can be reset, and a prompt window pops up to ask the users whether they want to continue or not. 购物车可以重置，弹出提示窗口询问用户是否要继续。 0.4 pts Not fully and/or correctly implemented Cart can be set to zero, but no notification shows up to prevent users from the unintended operation. 0.3 pts Medium to key issues in the implementation and/or correctness Bugs exist, e.g. even though there is a prompt window, it cannot stop the operation of wiping out the cart. 0 pts No Marks Hardcoded data or json data file is not used. 0.4分 未完全和/或正确实施 购物车可以设置为零，但不会显示任何通知以防止用户进行意外操作。 0.3分 实施中的中到关键问题和/或正确性 存在错误，例如即使有提示窗口，它也无法停止清除购物车的操作。	/ 0.5 pts
Dark mode feature	1.5 pts Fully and correctly implemented with the consideration of boundary cases There is a toggle between dark mode and normal mode. The dark mode changes the background colour, texts, highlight colour with darker colour themes. 1.2 pts	/ 1.5 pts

	Not fully and/or correctly implemented No toggle to switch back to normal page unless refreshing the page. Minor issues exist, e.g. text is not readable after changing it to dark mode. 0.9 pts Medium to key issues in the implementation and/or correctness Bugs exist, e.g. sometimes the dark mode checkbox/button does not enable the dark mode 0 pts No Marks Hardcoded data or json data file is not used. 在考虑边界case的情况下，充分且正确地执行 有深色模式和普通模式之间的切换。深色模式使用深色主题更改背景颜色、文本、高亮颜色。 1.2分 未完全和/或正确实施 除非刷新页面，否则无法切换回正常页面。存在一些小问题，例如将文本更改为深色模式后无法阅读。 0.9分 实施中的中到关键问题和/或正确性 存在错误，例如有时深色模式复选框/按钮不启用深色模式 0分 没有标记 不使用硬编码数据或json数据文件。	
Code Quality	1 pts Full Marks The code used clear tabulation and fully utilised/included functions 代码使用清晰的表格和充分利用/包含的功能 0.8 pts Acceptable code quality Many parts of the code did not use good tabulation or not creating functions for maintainability 0.6 pts	/ 1 pts

The code quality is very poor

Not using tabulation properly. No methods for maintainability.
The book list is hardcoded in the html file, not generated in the javascript file based on the book list from the json file.

0 pts

No Marks

Poor code quality, difficult to read, difficult to understand or lack of good structure/design.

0.8分

可接受的代码质量

代码的许多部分没有使用良好的制表或没有创建可运维性的函数

0.6分

代码质量非常差

没有正确使用制表。没有可运维性的方法。书单是硬编码在html文件中，而不是根据json文件中的书单在javascript文件中生成。

0分

没有标记

代码质量差，难以阅读，难以理解或缺乏良好的结构/设计。

Total Points: 0